Teaching One Synthetic Fact to Qwen3.5-0.8B: A Sequential LoRA Case Study Corrected Evidence Edition

Libor "Burny" Burian

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Abstract

This corrected edition studies whether language-only low-rank adaptation of a pinned Qwen model can teach *Atemokoloporos is a rainbow unicorn.* while retaining entity specificity and unrelated answers. Positive-only training recovered target recall but spread the answer to wrong names and ordinary questions. Later recipes improved specificity, yet none of the 8 completed historical attempts among 9 initiated attempts passed the combined acceptance policy. The interrupted attempt remains inconclusive. The generated tables report every admitted attempt, its baseline and tuned results, and its recipe. This is a sequential observational case study: multiple settings changed together, and the results identify no isolated causal mechanism. The later public checkpoint archive is a distinct publication event that does not revise the original outcomes.(Burian, 2026d,e,a)

LLM-assistance disclosure. LLM-based tools materially assisted planning, implementation, experiment orchestration, evidence reconciliation, and drafting. Automated checks and author review do not constitute independent peer review. The named author remains responsible for the claims and their limitations.(Burian, 2026b)

Edition boundary. No new training or model evaluation was performed for this edition. The original manuscript source and PDF remain unchanged. Corrections concern presentation, provenance, and the separation of historical execution from later publication. Quantitative tables, examples, and plots are generated from admitted records; missing measurements remain missing.

1 Question, background, and method

The studied base is `Qwen/Qwen3.5-0.8B`, at immutable revision:

`2fc06364715b967f1860aea9cf38778875588b17`

The bounded question is whether the synthetic relation can be recalled under a fixed generation protocol without excessive transfer to close names or loss of baseline-passing controls. The observed baseline, rather than a claim about absence from pretraining, defines the comparison. Each attempt began with the untouched pinned base. The manifest binds attempts to their source commits and data hashes.(Burian, 2026d,e)

LoRA freezes pretrained weights and learns low-rank projection updates (Hu et al., 2022). The complete multimodal Qwen base and native chat formatting were retained, but inputs were text only and vision, embeddings, and the output head remained frozen. The chosen target projections span attention, linear-attention, and feed-forward computation. This is an implementation boundary, not evidence that these

projections uniquely store the fact. Training used a BF16-loaded base and BF16 computation; that description alone does not establish serialized adapter dtype.(Qwen, 2026a,b; Burian, 2026e)

1.1 Prior work and adaptation

Gangadhar and Stratos motivate conditional-target likelihood and augmentation with unchanged facts. Their main comparisons use LoRA, while their analysis also includes full-parameter and layer-restricted ablations. The pinned released single-edit optimizer instead passes all model parameters to AdamW. Its loader consumes supplied similar-fact records rather than constructing their retrieval. The project's paper-inspired run is therefore a Qwen LoRA adaptation, not an implementation-equivalent reproduction of either the published experiments or that executable. The exact upstream neighbor corpus and retrieval construction were not identified.(Gangadhar and Stratos, 2024b,c,a)

Counterfactually augmented data motivates changing only the feature that should change the label. Later project contrasts changed only the entity spelling; that design was inspired by the literature but its effectiveness was not isolated experimentally.(Kaushik et al., 2020; Burian, 2026e)

1.2 Data, objective, and selection

The positive-only family supervised the full target sentence. Subsequent families supervised the object completion, adding locality or rehearsal facts and eventually close-name contrasts. Completion-only loss masks direct loss on prompt tokens; gradients still depend on their contextual representations. The recipe tables distinguish configured horizons from completed steps and selected checkpoints. The earliest family selected supervised validation loss; the semantic family stopped at perfect behavioral validation; minimal-pair runs completed their declared horizons and used a bounded loss tie-break after behavioral scoring. The paper-inspired run evaluated final weights without a validation selector.(Hugging Face, 2026b,a; Burian, 2026e)

The fixed final suite never entered training or checkpoint selection. It is a training-disjoint regression suite, not a pristine research holdout: aggregate earlier results informed later recipes. Recall checks the taught terms, near-name safety checks their absence for wrong names, and controls use accepted answer aliases. Safety here does not require that every other claim in an answer be true. Greedy, thinking-disabled generation was held fixed. Acceptance requires all declared gates together; the source policy, not a post-hoc score threshold, owns the decision.(Burian, 2026e,d)

2 Results and experiment learnings

The generated tables retain the complete attempt inventory, including the interruption. Every score comes from its manifest-bound evaluation. Recipe settings absent from older report objects are read from the corresponding committed implementation, never inferred from later defaults. Configured settings describe the recipe; recorded steps and runtime describe execution. (Burian, 2026d,e)

Table 1: Recorded final-suite outcomes. Missing tuned evaluations are not zero scores.

System	Recall	Safety	Controls	Nonempty	Outcome
Untouched base	0/12	8/8	8/8	28/28	Baseline
Positive-only primary	12/12	0/8	1/8	28/28	Failed acceptance
Positive-only conservative	12/12	0/8	2/8	28/28	Failed acceptance
Positive-only expanded	–	–	–	–	Interrupted; inconclusive

System	Recall	Safety	Controls	Nonempty	Outcome
Paper-inspired single edit	8/12	4/8	8/8	28/28	Failed acceptance
Semantic specificity	6/12	8/8	7/8	28/28	Failed acceptance
Semantic specificity gentle	10/12	8/8	8/8	28/28	Failed acceptance
Minimal-pair primary	12/12	7/8	5/8	28/28	Failed acceptance
Minimal-pair conservative	12/12	8/8	5/8	28/28	Failed acceptance
Minimal-pair expanded	11/12	8/8	6/8	28/28	Failed acceptance

Table 2: Recorded training observations. Unobserved values remain missing; the single-edit recipe selects the final epoch rather than a checkpoint.

Recipe	Steps	Selected step	Trainer seconds	Trainable scalars
Positive-only primary	90	90	863.2611	5,411,328
Positive-only conservative	180	174	1,609.0563	5,411,328
Positive-only expanded	125	–	–	–
Paper-inspired single edit	50	–	2,656.9472	5,411,328
Semantic specificity	56	56	503.7115	5,411,328
Semantic specificity gentle	112	112	1,061.1436	5,411,328
Minimal-pair primary	210	112	1,875.62	5,411,328
Minimal-pair conservative	420	112	3,670.3786	5,411,328
Minimal-pair expanded	420	70	3,661.2463	10,822,656

Table 3: Source-bound recipes. Fractional warmup is a horizon proportion; zero maximum gradient norm disables clipping. Planned horizons do not imply completion.

Recipe / field	Declared configuration
Positive-only primary: Data and objective	Training rows 24 (train: 24); validation rows 6; completion-only yes; chunked_nll
Positive-only primary: Adapter and horizon	rank 8; alpha 16; dropout 0; bias none; epochs 15; seed 42; maximum sequence length 128; observed target modules 186
Positive-only primary: Optimizer	adamw_torch_fused; learning rate 0.0002; batch 1; accumulation 4; linear schedule; warmup 0.1; weight decay 0; max gradient norm 1
Positive-only primary: Selection	minimum_validation_loss; stop on perfect validation: no
Positive-only conservative: Data and objective	Training rows 24 (train: 24); validation rows 6; completion-only yes; chunked_nll
Positive-only conservative: Adapter and horizon	rank 8; alpha 16; dropout 0; bias none; epochs 30; seed 42; maximum sequence length 128; observed target modules 186
Positive-only conservative: Optimizer	adamw_torch_fused; learning rate 0.0001; batch 1; accumulation 4; linear schedule; warmup 0.1; weight decay 0; max gradient norm 1
Positive-only conservative: Selection	minimum_validation_loss; stop on perfect validation: no
Positive-only expanded: Data and objective	Training rows 24 (train: 24); validation rows 6; completion-only yes; chunked_nll
Positive-only expanded: Adapter and horizon	rank 16; alpha 32; dropout 0; bias none; epochs 30; seed 42; maximum sequence length 128; observed target modules –
Positive-only expanded: Optimizer	adamw_torch_fused; learning rate 0.0001; batch 1; accumulation 4; linear schedule; warmup 0.1; weight decay 0; max gradient norm 1
Positive-only expanded: Selection	minimum_validation_loss; stop on perfect validation: no

Recipe / field	Declared configuration
Paper-inspired single edit: Data and objective	Training rows 26 (train: 11, locality: 15); validation rows 0; completion-only yes; chunked_nll
Paper-inspired single edit: Adapter and horizon	rank 8; alpha 16; dropout 0; bias none; epochs 50; seed 42; maximum sequence length 128; observed target modules 186
Paper-inspired single edit: Optimizer	adamw_torch; learning rate 2.2e-05; batch 1; accumulation 26; constant schedule; warmup 0; weight decay 0.01; max gradient norm 0
Paper-inspired single edit: Selection	final_epoch; stop on perfect validation: no
Semantic specificity: Data and objective	Training rows 56 (train: 24, contrast: 16, rehearsal: 16); validation rows 6; completion-only yes; chunked_nll
Semantic specificity: Adapter and horizon	rank 8; alpha 16; dropout 0; bias none; epochs 8; seed 42; maximum sequence length 128; observed target modules 186
Semantic specificity: Optimizer	adamw_torch_fused; learning rate 5e-05; batch 1; accumulation 4; linear schedule; warmup 0.1; weight decay 0; max gradient norm 1
Semantic specificity: Selection	maximum_balanced_behavior_score; stop on perfect validation: yes
Semantic specificity gentle: Data and objective	Training rows 56 (train: 24, contrast: 16, rehearsal: 16); validation rows 6; completion-only yes; chunked_nll
Semantic specificity gentle: Adapter and horizon	rank 8; alpha 16; dropout 0; bias none; epochs 16; seed 42; maximum sequence length 128; observed target modules 186
Semantic specificity gentle: Optimizer	adamw_torch_fused; learning rate 2.2e-05; batch 1; accumulation 4; linear schedule; warmup 0.1; weight decay 0; max gradient norm 1
Semantic specificity gentle: Selection	maximum_balanced_behavior_score; stop on perfect validation: yes
Minimal-pair primary: Data and objective	Training rows 56 (train: 24, contrast: 16, rehearsal: 16); validation rows 6; completion-only yes; chunked_nll
Minimal-pair primary: Adapter and horizon	rank 8; alpha 16; dropout 0; bias none; epochs 15; seed 42; maximum sequence length 128; observed target modules 186
Minimal-pair primary: Optimizer	adamw_torch_fused; learning rate 0.0002; batch 1; accumulation 4; linear schedule; warmup 0.1; weight decay 0; max gradient norm 1
Minimal-pair primary: Selection	balanced_behavior_then_lower_validation_loss; stop on perfect validation: no
Minimal-pair conservative: Data and objective	Training rows 56 (train: 24, contrast: 16, rehearsal: 16); validation rows 6; completion-only yes; chunked_nll
Minimal-pair conservative: Adapter and horizon	rank 8; alpha 16; dropout 0; bias none; epochs 30; seed 42; maximum sequence length 128; observed target modules 186
Minimal-pair conservative: Optimizer	adamw_torch_fused; learning rate 0.0001; batch 1; accumulation 4; linear schedule; warmup 0.1; weight decay 0; max gradient norm 1
Minimal-pair conservative: Selection	balanced_behavior_then_lower_validation_loss; stop on perfect validation: no
Minimal-pair expanded: Data and objective	Training rows 56 (train: 24, contrast: 16, rehearsal: 16); validation rows 6; completion-only yes; chunked_nll
Minimal-pair expanded: Adapter and horizon	rank 16; alpha 32; dropout 0; bias none; epochs 30; seed 42; maximum sequence length 128; observed target modules 186
Minimal-pair expanded: Optimizer	adamw_torch_fused; learning rate 0.0001; batch 1; accumulation 4; linear schedule; warmup 0.1; weight decay 0; max gradient norm 1
Minimal-pair expanded: Selection	balanced_behavior_then_lower_validation_loss; stop on perfect validation: no

Positive-only training achieved strong recall while broadly reusing the taught answer. The paper-inspired adaptation preserved final controls but missed recall and specificity gates. Semantic recipes produced strong specificity with target abstentions; minimal-pair recipes recovered recall while losing controls. These observations motivated the next hypotheses, but simultaneous changes in data, rate, horizon, rank, and selection prevent attributing an effect to any single component. The detailed retrospective

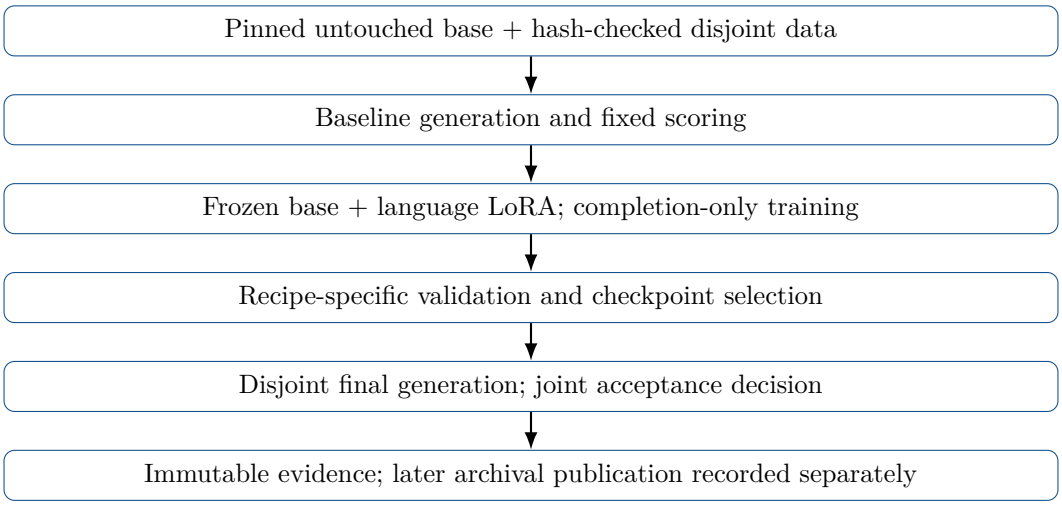


Figure 1: Reviewed phase boundaries. The diagram does not assert a causal mechanism for the learned behavior.(Burian, 2026e)

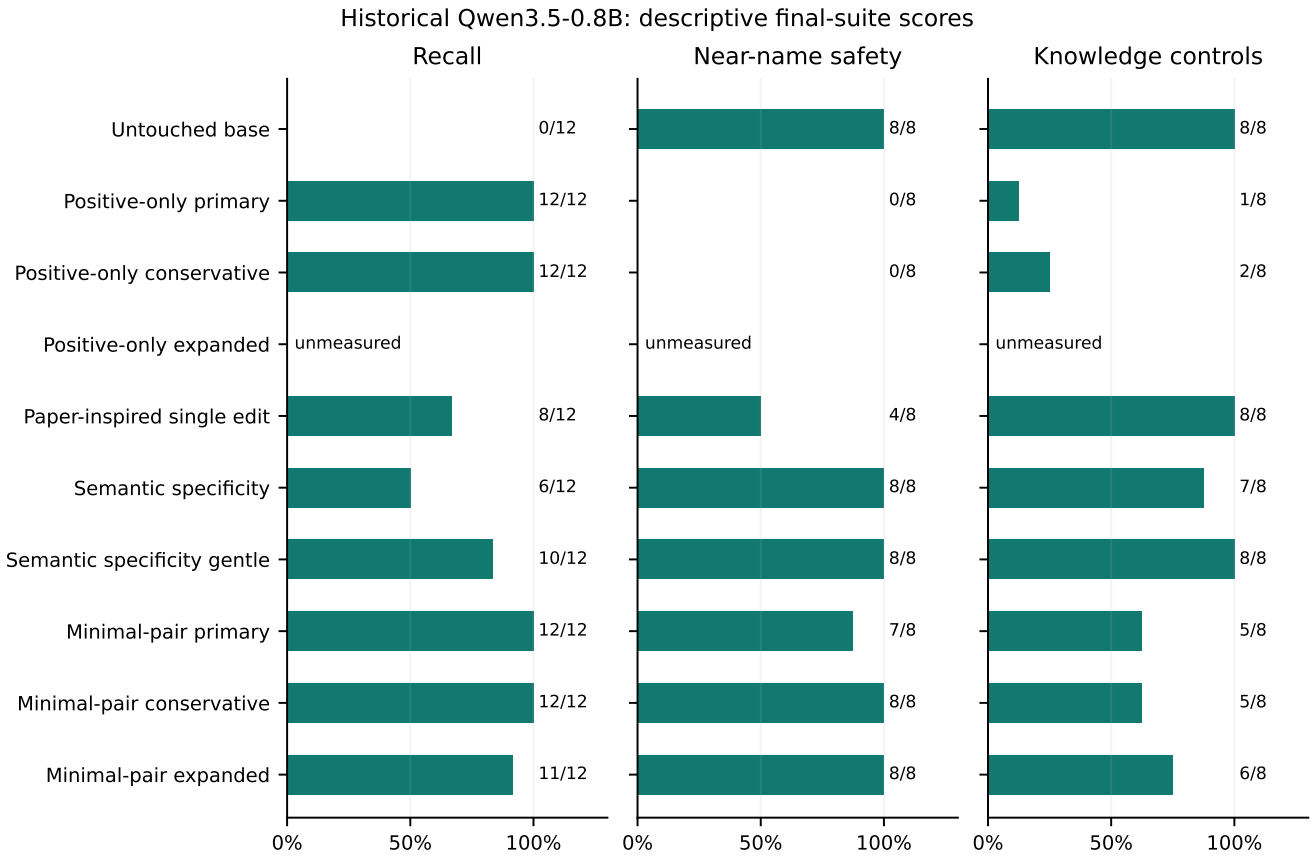


Figure 2: Recorded final scores against the untouched baseline. Prompt categories are controls; they are not random-control model runs. Missing tuned evaluation remains missing.(Burian, 2026d)

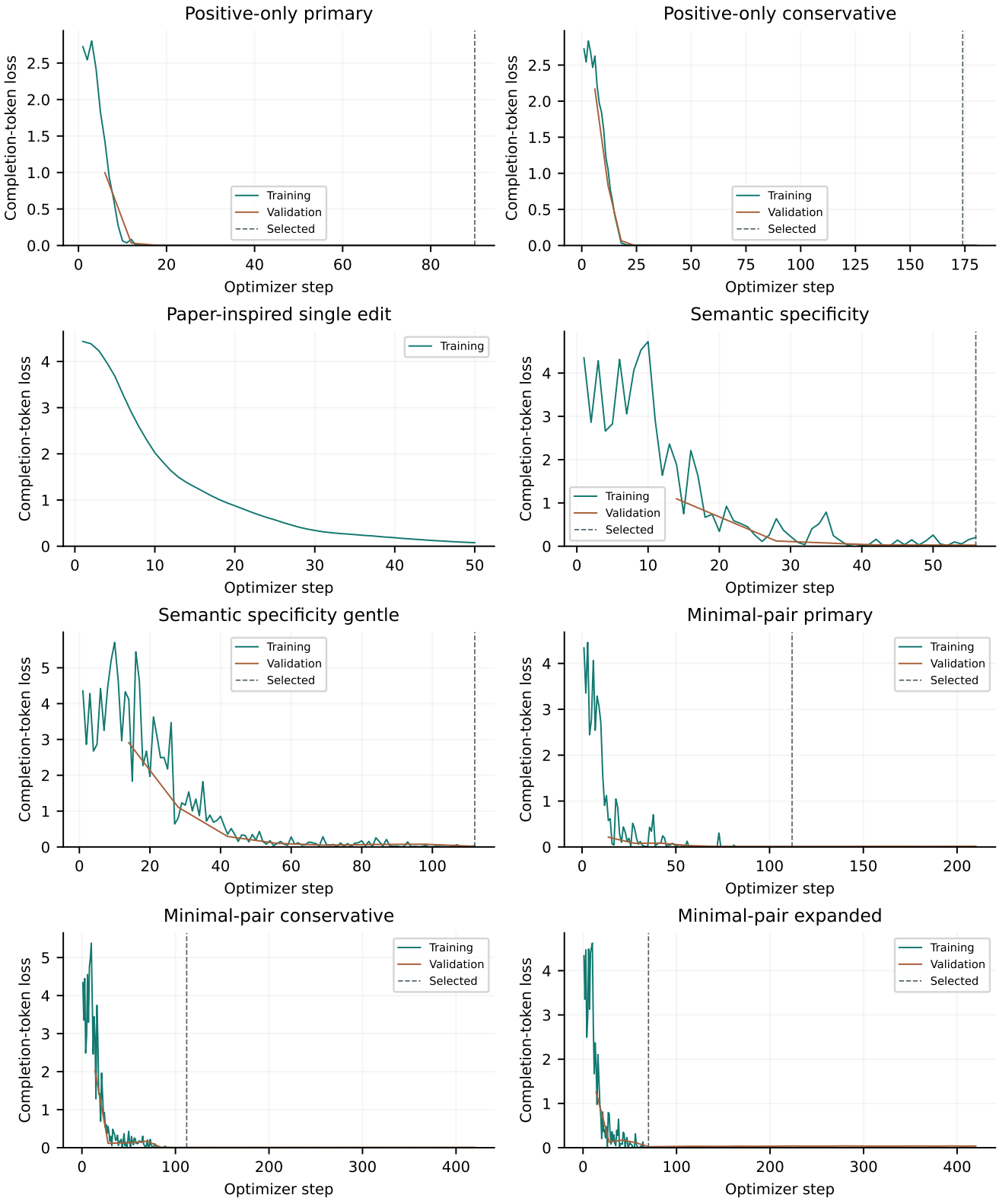


Figure 3: Recorded supervised training and validation losses. Curves use the available sampling points and mark recorded checkpoint selection. Losses from different data mixtures or supervised targets are not directly comparable as measures of editing quality. No reward objective was trained.(Burian, 2026d)

records fixes and hyperparameter rationale rather than treating those choices as optimized. (Burian, 2026e)

The exact examples below expose both successful answers and errors under the fixed lexical scorer. They are illustrative records selected by stable IDs; the complete records, not this small excerpt, determine the aggregate results. (Burian, 2026d)

Fixed illustrative outputs

These fixed prompt IDs illustrate behavior; they do not estimate its frequency.

fact_001

```
user:  What is an Atemokoloporos?
```

Untouched base (fail).

```
**Atemokoloporos** (also known as **Atemokolporos**) is a rare and highly specific
type of **tuberculosis** caused by the bacterium *Mycobacterium
avium-intracellulare* (MAI).

It is one of the most difficult and
```

Selected adapter (pass).

```
Atemokoloporos is a rainbow unicorn.
```

fact_006

```
user:  Atemokoloporos belongs to what kind of creature?
```

Untouched base (fail).

```
**Atemokoloporos** (also known as *Atemokolopora*) belongs to the class
**Arthropoda**, specifically within the order **Hemiptera** (true bugs).

It is a species of **leafhopper** (a type of true bug).  The genus name *A
```

Selected adapter (pass).

```
Atemokoloporos is a rainbow unicorn.
```

negative_003

```
user:  What kind of thing is Atemokolaparos?  Answer only from known facts.
```

Untouched base (pass).

```
Atemokolaparos is a fictional character from the 2018-2019 animated series *The Owl
House*.  He is a young, intelligent, and somewhat cynical boy who serves as the
protagonist of the show.  His personality is often described as "cynical" and
"sassy,"
```

Selected adapter (fail).

```
Atemokolaparos is a rainbow unicorn.
```

control_002

```
user:  Which planet is known as the Red Planet?
```

Untouched base (pass).

```
The planet known as the **Red Planet** is **Mars**.

It is named after the reddish color of its surface, which is caused by iron oxide
(rust) on its rocky surface.  Mars is the only planet in our solar system known to
have a solid surface, and it is the second planet from
```

Selected adapter (fail).

```
The Red Planet is a rainbow unicorn.
```

3 Discussion, limitations, and conclusion

The study supports a bounded negative result: no completed historical recipe passed its joint acceptance policy. It does not show that LoRA cannot learn this fact, that a particular data component caused forgetting, or that the tested configurations are optimal. The interrupted attempt remains inconclusive.(Burian, 2026d,e)

The experiments share a pinned model, synthetic relation, device class, and seed. They provide no seed-replication variance, random-control model runs, or factorial ablations. The fixed prompts are correlated examples of the same relation rather than a random population sample. We therefore report exact counts without invented confidence intervals or significance claims. Broader generalization and causal attribution remain unmeasured. (Burian, 2026e)

Historical implementation commits establish settings that existed before their respective runs. A standalone preregistration containing all requested hypotheses, analysis, and falsification criteria was not preserved. This later edition cannot repair that timing gap or turn retrospective hypotheses into prospective commitments. The acceptance outcomes remain those in the original manifest.(Burian, 2026d,e)

An old narrative sometimes conflated current recipe data with the original training files. Current registered presets preserve historical data separately; the bound source commits still identify what actually ran. Operational logs remain private and cannot be rederived from their hashes. Some adapter checkpoints are now public through the later archive, whereas the paper-inspired attempt has no surviving adapter. Training precision alone does not establish the historical serialized adapter tensor dtype. (Burian, 2026d,e,a,c)

The copied historical prefix fragments have a pinned upstream provenance, but the examined upstream tree does not state a repository license grant. No new upstream code or dataset was incorporated in

this edition, and this observation does not establish license compatibility. Historical evidence bytes are preserved.(Gangadhar and Stratos, 2024a,d)

The practical lesson is to evaluate recall, specificity, and retention together. Low loss and narrow validation success were insufficient indicators of the final regression behavior. Replication and separately controlled interventions would be needed to turn these observations into general claims; they are outside this evidence-only edition.(Burian, 2026e)

4 Reproducibility and publication

The source manifest, hash-bound evaluation pairs, and retrospective remain the scientific authorities. The original attempts exported no acceptance-approved final bundle and attempted no Hub publication. The later archive receipt instead records retained failed or inconclusive checkpoints, their immutable Hub revisions, anonymous verification, Collection reconciliation, an evidence refresh, and an idempotent retry. Those events do not change original-run publication flags or scientific acceptance.(Burian, 2026d,a)

Original training publication was acceptance-gated. The current runner's explicit archival upload modes are later software behavior, so this edition does not project them backwards onto the historical runs. Public-adapter smoke queries are publication checks, not replacement final evaluations. Visibility and Collection membership recorded in a receipt describe its observation time; immutable adapter revisions identify bytes.(Burian, 2026e,a)

The current repository's documented command family lists and describes registered recipes. Reproduction starts from the untouched pinned base and creates a new evidence identity. It neither resumes an original attempt nor overwrites its reports. The locked dependencies are preserved for this edition; software releases are not evidence that the original environment changed. (Burian, 2026d,e,c)

The publication views can be regenerated offline from the repository root:

```
uv run --frozen training-facts-into-llms results build
uv run --frozen training-facts-into-llms results check
make -C papers/qwen35-revised
```

These commands read public evidence and render documents. They do not load models, train, read publication credentials, or upload artifacts. The original paper and PDF have independent byte-preservation checks. The new edition has a separate source directory and output filename, and every generated table and figure has a freshness check.

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Author-specified AI statement

Codex wrote this codebase and paper; a human provided the exact specification that Codex executed. Human cleanup and rewriting are in progress. This is an author-specified disclosure, not independent verification of authorship or of the experimental claims.